BJT and MOSFETS

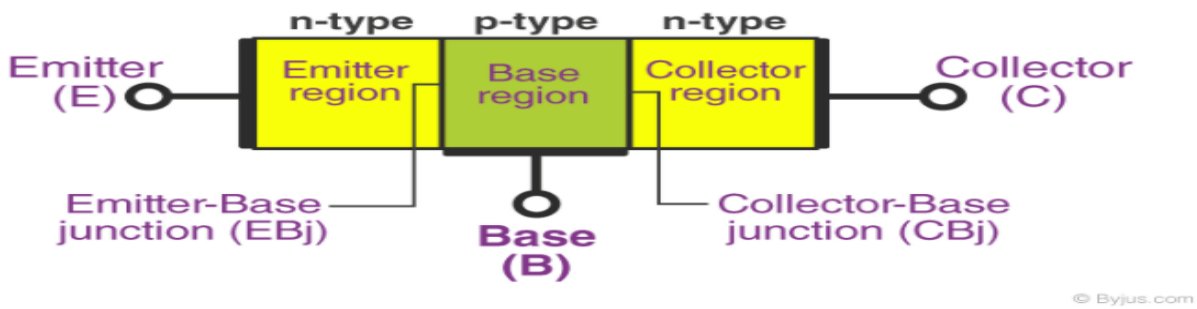
Bipolar Junction Transistor:

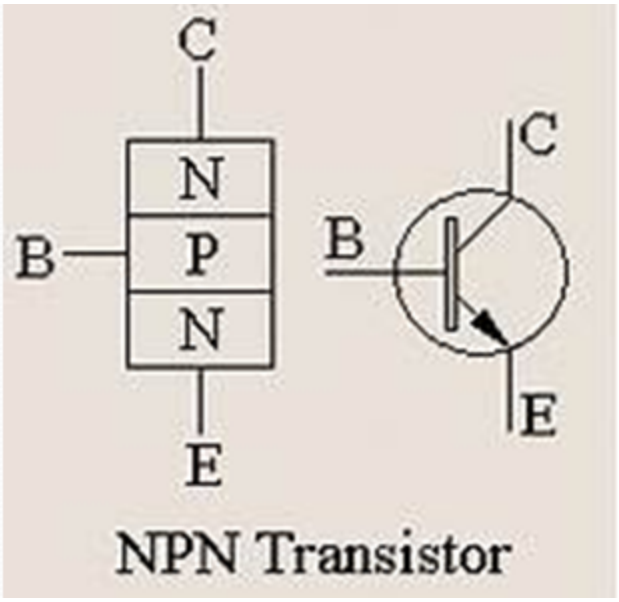
A bipolar junction transistor is a three-terminal semiconductor device that consists of two p-n junctions which are able to amplify or magnify a signal. It is a **current controlled device**. The three terminals of the BJT are the base, the collector, and the emitter. A signal of a small amplitude applied to the base is available in the amplified form at the collector of the transistor. This is the amplification provided by the BJT. BJT is a semiconductor device that is constructed with 3 doped semiconductor Regions i.e. **Base, Collector & Emitter** separated by 2 p-n Junctions. Bipolar transistors are manufactured in two types, **PNP** and **NPN**, and are available as separate components, usually in large quantities. The prime use or function of this type of transistor is to amplify current. This makes them useful as switches or amplifiers. They have a wide application in electronic devices like mobile phones, televisions, radio transmitters, and industrial control.

Types of BJT:

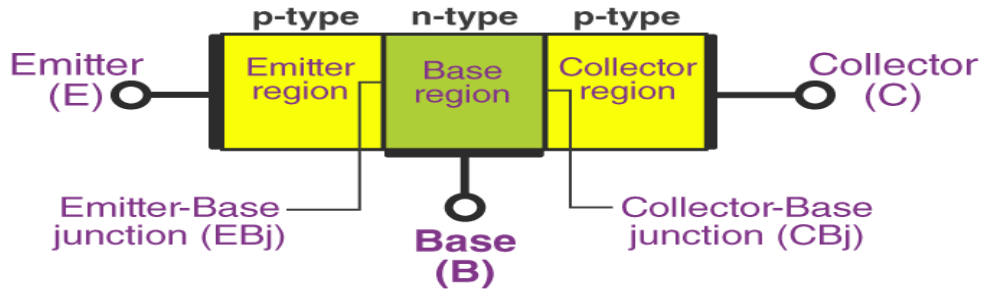
NPN transistor:

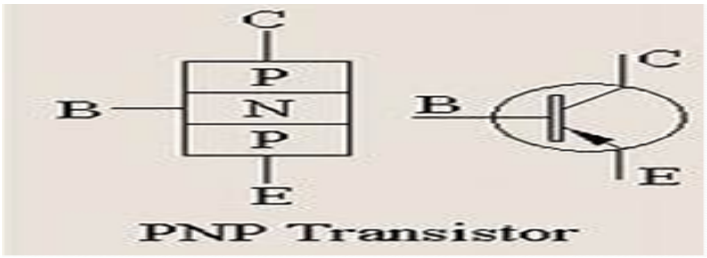
In NPN BJT, p-type semiconductor is sandwiched between the two n-type semiconductors. The two n-type semiconductors act as emitter and collector respectively while the p-type semiconductor acts as a base. structure and symbol is shown in fig below





PNP Transistor: In PNP BJT, the n-type semiconductor is sandwiched between the two p-type semiconductors. The two p-type semiconductors act as emitter and collector respectively while the n-type semiconductor acts as a base. Structure and symbol is shown in the figure below.





Operation of BJT:

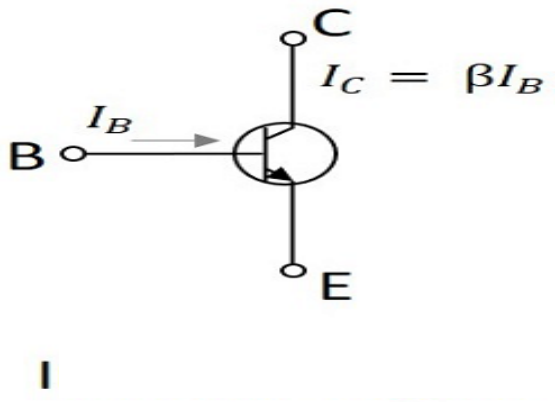
There are three operating regions of a bipolar junction transistor:

- Active region: The region in which the transistors operate as an amplifier.
- Saturation region: The region in which the transistor is fully on and operates as a switch such that collector current is equal to the saturation current.
- Cut-off region: The region in which the transistor is fully off and collector current is equal to zero.

EMITTER JUNCTION	COLLECTOR JUNCTION	REGION OF OPERATION
Forward biased	Forward biased	Saturation region
Forward biased	Reverse biased	Active region
Reverse biased	Reverse biased	Cut off region

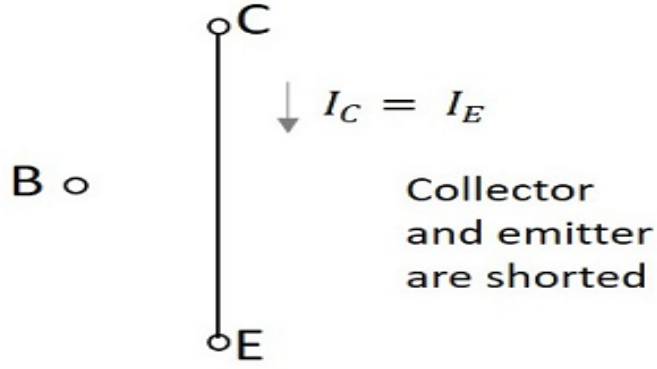
Active Region

Transistor act as amplifier.



Saturation Region

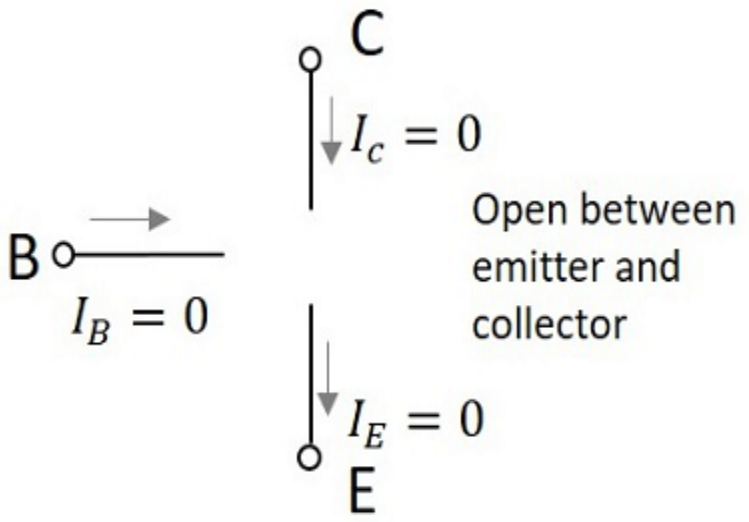
This is the region in which transistor tends to behave as a closed switch. The transistor has the effect of its collector and Emitter being shorted. The collector and Emitter currents are maximum in this mode of operation and collector current is equal to the saturation current.



In Saturated region

Cut off Region

This is the region in which transistor tends to behave as an open switch. The transistor has the effect of its collector and base being opened. The collector, emitter and base currents are all zero in this mode of operation.



In Cutoff region

Transistor application as Switch

Transistor as switch

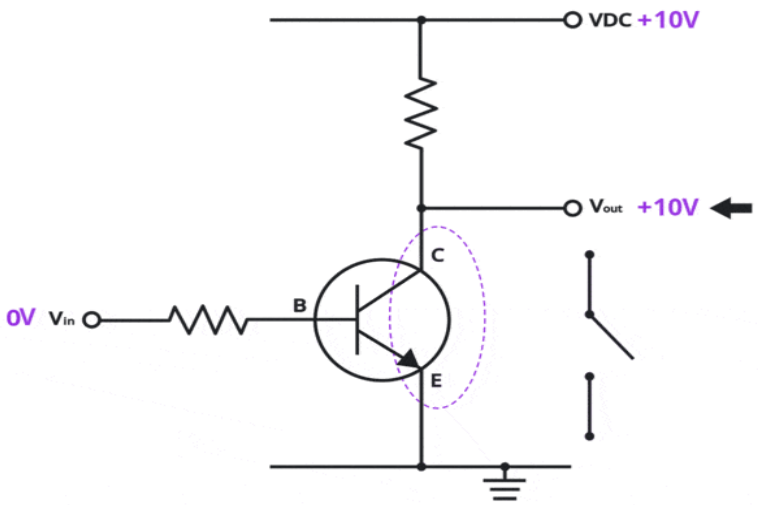
When we are using the transistor as switch we deal

with only **saturation region** and **cut off region**.

In the cut off region the base voltage is less than 0.7 V so the base current is also negligible. In the transistor, the collector current is directly proportional to the base current. So the collector current Ic is also negligible. But the collector to emitter voltage V_{CE} is maximum. Which creates a large depletion region and no current flows through the transistor. So this region is called as cut off region.

In this mode the maximum base current is applied that results with high collector current and lower collector to emitter voltage. And this results a lower depletion region layer. By this the large amount of current can flow through the transistor. The conditions for the transistor

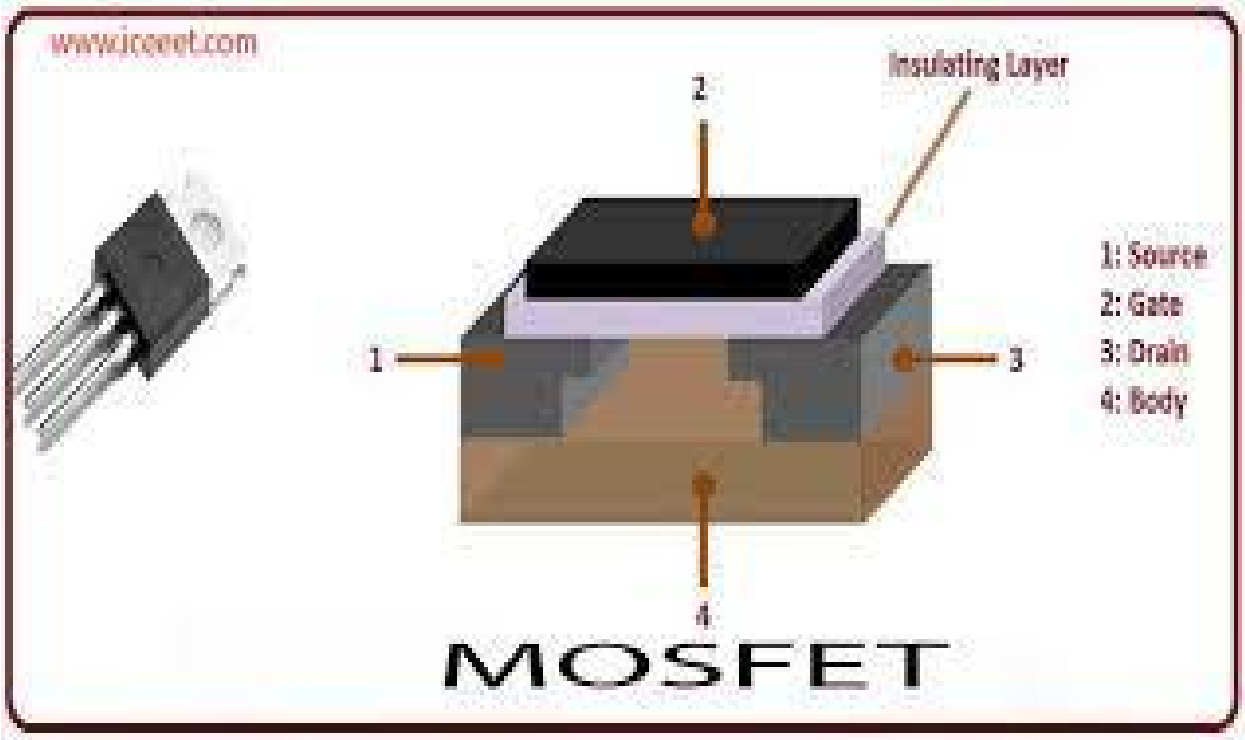
in the saturation region are **V_b > 0.7 V, Collector Current (Ic) = maximum, Collector to Emitter Voltage (V_{CE}) = minimum.**



Metal Oxide Semiconductor Field Effect Transistor

The MOSFET (Metal Oxide Semiconductor Field Effect Transistor) is a semiconductor device that is widely used for switching purposes and for the amplification of electronic signals in electronic devices.

A MOSFET is a four-terminal device having source(S), gate (G), drain (D) and body (B) terminals. In general, The body of the MOSFET is in connection with the source terminal thus forming a three-terminal device such as a field-effect transistor. MOSFET is generally considered as a transistor and employed in both the analog and digital circuits. The general structure of this device is as below :

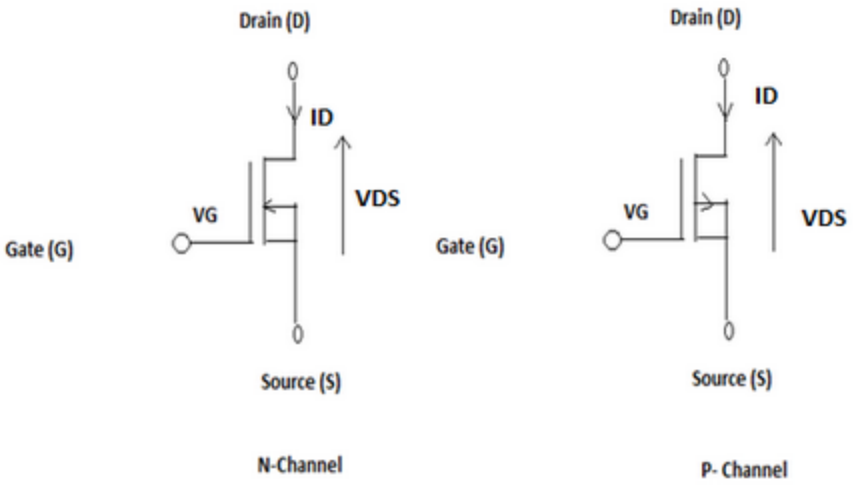


From the above MOSFET structure, the functionality of MOSFET depends on the electrical variations happening in the channel width along with the flow of carriers (either holes or electrons). The charge carriers enter into the channel through the source terminal and exit via the drain.

The width of the channel is controlled by the voltage on an electrode which is called the gate and it is located in between the source and the drain. It is insulated from the channel near an extremely thin layer of metal oxide. The MOS capacity that exists in the device is the crucial section where the entire operation is across this.

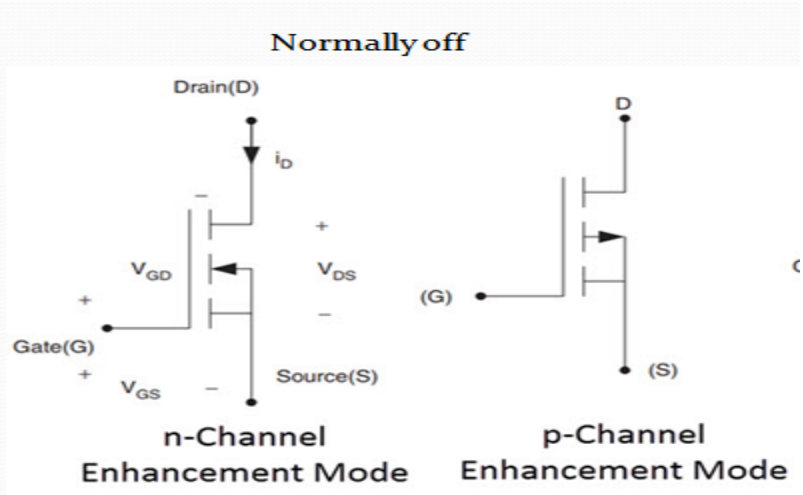
A MOSFET can function in two ways

- Depletion Mode: The transistor requires the Gate-Source voltage (VGS) to switch the device “OFF”. The depletion-mode MOSFET is equivalent to a “Normally Closed” switch.



- Enhancement Mode

The transistor requires a Gate-Source voltage(VGS) to switch the device “ON”. The enhancement-mode MOSFET is equivalent to a “Normally Open” switch.



Regions of operation of MOSFET

MOSFET has three regions of operation. They are:

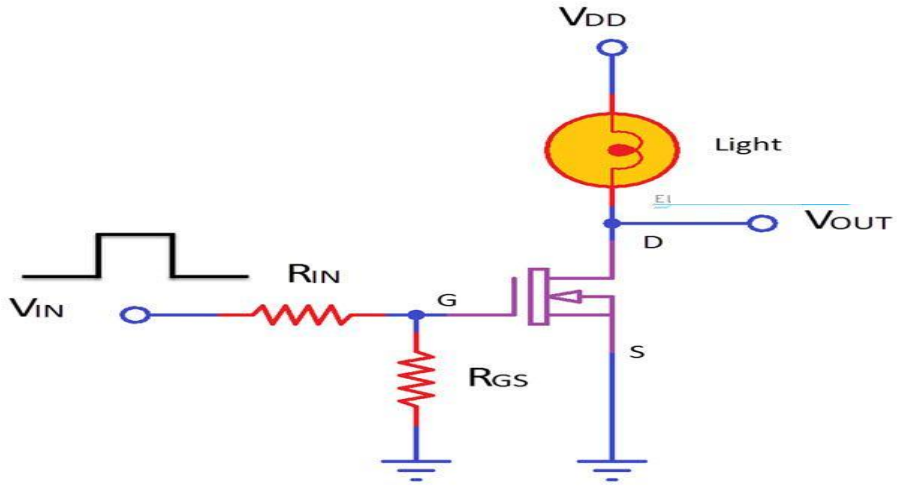
- Cut-off Region
- Linear (or Triode) Region
- Saturation Region

A MOSFET operates in cut-off region when $V_{GS} < V_{TH}$. In this region, the MOSFET is in OFF state as there is no channel induced between drain and source.

For the channel to be induced and MOSFET to operate in either linear or saturation region, $V_{GS} > V_{TH}$. The Gate – Drain bias voltage V_{GD} will determine whether the MOSFET is in linear or saturation region. In both these regions, the MOSFET is in ON state but the difference is in linear region, the channel is continuous and the drain current is proportional to the resistance of the channel.

Coming to saturation region, as $V_{DS} > V_{GS} – V_{TH}$, the channel pinches off i.e., it broadens resulting in a constant Drain Current.

Application of MOSFET as switch:



This is a simple circuit, where an N-Channel Enhancement mode MOSFET will turn ON or OFF a light. In order to operate a MOSFET as a switch, it must be operated in cut-off and linear (or triode) region.

Assume the device is initially OFF. The voltage across Gate and Source i.e., V_{GS} is made appropriately positive (technically speaking, $V_{GS} > V_{TH}$), the MOSFET enters linear region and the switch is ON. This makes the Light to turn ON.

If the input Gate voltage is 0V (or technically < V_{TH}), the MOSFET enters cut-off state and turns off. This in turn will make the light to turn OFF.

Difference between BJT and MOSFET

	BJT	Mosfet
Type	PNP or NPN	N-type or P-Type
Device	Current Controlled Device	Voltage Controlled Device
Output controlled	By controlling base current	By controlling gate voltage
Temperature coefficient	Negative temperature coefficient	Positive temperature coefficient
Input Impedance	Low	High
Switching Frequency	Low	High
Preferred	Low Current Application	Hight Power Application